

Locating Where Extreme-Score Detectors Fail: A Layered Evaluation for Predictive Maintenance on Public Industrial Data

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Abstract

Whether clustered extreme scores become useful maintenance alarms is a question about *where* a detector fails, not only whether it does. This paper builds a layered evaluation that separates a detector into score, threshold, extreme clusters, alarm episodes and matched events, and applies it to public industrial monitoring data. A Monte Carlo study over fourteen process families at 500 replications per cell shows the ranking of six extremal-index estimators inverting between regimes: runs and K-gaps are among the worst calibrated for an independent process and the least biased of the six once extremes genuinely cluster, so a reader seeing only near-independent processes would reject the estimators that work. Nineteen methods are then evaluated on two independent held-out compressor failures under one shared alarm policy. Measuring how well each score ranks the hour before onset against the rest of the test split, most separate well above chance on both datasets, and 25 of the 38 dataset-method rows fail only after alarm conversion: the registered recurrence score matches both failures, at 2770 and 893 alarm episodes and event precision below 0.12%. A circular-shift null, which preserves each detector’s episode structure while breaking its alignment to the failure, shows that detectors alarming this often match with probability above 0.99 regardless, so recall alone carries almost no information at these rates. A single-sensor threshold remains the strongest simple baseline, and target regions transfer directionally and only on the five of seven features with cross-dataset distributional support. One configuration is different: on MetroPT2 the failure-prototype region raises 3 alarm episodes, matches the failure, and reaches 0.12 false alarms per operating day. Its chance-match probability is 0.17 and those three episodes hold the system under warning for 16% of the test period, so the result is suggestive rather than established. On two failures, no evaluated method is shown to deliver reliable early warning, and none is shown to be incapable of it.

1 Introduction

Industrial maintenance alarms are useful only when they map to the decision unit that operators and engineers actually face. A pointwise anomaly score may flag thousands of telemetry samples around one physical failure, while the operator sees one alarm burden, one repair decision, and one opportunity for useful warning. The same difficulty appears outside compressor telemetry: a fleet repair record, a laboratory hydraulic cycle, and a semiconductor yield label are not exchangeable evidence units. Any empirical claim must therefore separate samples, exceedances, clusters, alarm episodes, failures, vehicles, cycles, and production examples.

Dynamical extreme-value theory offers a tempting route from recurrent machine states to early-warning logic. If an observable is large when the trajectory returns near a target set, then recurrence and periodicity can cluster extremes [7, 8]. In industrial monitoring, however, clustered extremes can also arise from operating regimes, preprocessing, smoothing, missingness, persistent shifts, or alarm-conversion rules. Noise can further weaken the connection between

deterministic recurrence and observed extremes [5]. The scientific question is therefore not whether an extremal-index diagnostic can be computed. The question is when such a diagnostic translates into useful episode-level warning.

This paper is framed as a negative-result study. It asks four research questions. First, does threshold and regime conditioning stabilise extremal-index diagnostics relative to global thresholding? Second, does declustering reduce duplicate alarm episodes without destroying event recall or warning lead time? Third, does the registered robust extreme-score diagnostic add operational value after episode conversion? Fourth, which mechanisms explain empirical failure: target-region misspecification, regime mixture, too few independent failures, non-extreme fault signatures, label ambiguity, calibration shift, or alarm conversion?

The evaluated workflow combines causal state construction, training-only target-region construction, regime-conditioned thresholds, extremal-index diagnostics, multivariate lag patterns, alarm merging, event matching, baseline provenance, and dataset-specific condition diagnostics. The interpretation is deliberately conservative so that negative and non-estimable outcomes remain visible.

The contribution is narrow. The study shows that dynamical-EVT-inspired diagnostics can reveal dependence, threshold sensitivity, duplicate-alarm behavior, and calibration limits, but the current public-data evidence does not show reliable early warning. MetroPT and MetroPT2 are the primary event-level case studies; SCANIA Component X is an external vehicle-level stress test; Hydraulic Systems and SECOM are supplementary condition-monitoring demonstrations. This separation is essential because the datasets support different estimands. The paper therefore reports what failed, where the evidence is insufficient, and which conditions would be required before making broader predictive-maintenance claims.

2 Related Work

2.1 Predictive-maintenance baselines

Predictive-maintenance models often begin with supervised classification, reconstruction error, distance-based anomaly detection, or streaming thresholds. Isolation Forest isolates anomalies by random partitioning and remains a standard unsupervised baseline [15]; autoencoders detect anomalies through reconstruction error under a learned nonlinear manifold [18]; offline and online change-point methods target shifts in the generating distribution rather than pointwise extremity [22]; and conformal prediction supplies distribution-free validity guarantees [25] that extend to sequential anomaly detection [13]. All five appear as baselines in the benchmark reported here. Their common output is a timestamped score rather than an episode-level maintenance decision, which is the gap this paper measures.

2.2 Dependence-aware extreme-value theory

Classical peaks-over-threshold modelling fits a generalised Pareto tail to exceedances over a high threshold [3], and the standard treatment covers threshold choice, return levels, and nonstationary extensions in which the tail parameters vary with covariates or time [2]. Independence of exceedances is the assumption that industrial telemetry violates most obviously. Leadbetter’s extremal index quantifies that violation as the reciprocal of the mean cluster size under local dependence [14], Ferro and Segers estimate it from inter-exceedance times [6], and the K-gaps estimator conditions on a declustering gap chosen from the data [20]. EVT-based streaming detectors such as SPOT adapt anomaly thresholds online using the same tail machinery [19]; they address threshold calibration, but not operating-regime mixture, duplicate alarms per failure, or the distinction between clustered observations and independent maintenance events.

In dynamical systems, hitting-time statistics connect returns to target neighborhoods with extreme-value laws [7]. Periodic points can generate an extremal index smaller than one, giving

a mathematical route from recurrence to clustered extremes [8]. The finite-sample industrial setting is less clean: noise may make deterministic recurrence appear closer to independence, and observed clustering can also arise from smoothing, missingness, persistence, or regime mixture [5]. Recent multivariate work extends dynamical EVT to cross-sectional dependence among vector-valued observables [1], which motivates the lag-pattern diagnostics used here.

2.3 Event-level evaluation and alarm burden

Pointwise precision and recall reward a detector for flagging many samples inside one long anomaly, so time-series evaluation requires range-aware definitions [21]. The point-adjust convention widely used in that literature is now known to inflate scores so severely that random scores can appear competitive [12]. This paper evaluates at the episode level for that reason, and reports the degenerate case explicitly: after alarm merging, a permanently-on detector attains precision and recall of one, so warning exposure is reported alongside episode counts throughout.

The operational cost of that burden is the subject of established alarm-management practice. Both the process-industry standard and the corresponding engineering guide specify alarm rates an operator can absorb and treat alarm rationalisation as a design activity rather than a post-hoc filter [4, 10]. The rates reported here, between 9 and 105 false alarms per operating day, are one to two orders of magnitude above those recommendations, which is the basis for the operational conclusion in Section 8.

2.4 Transferability and domain shift

Whether a detector calibrated on one machine transfers to another is rarely tested on the datasets used above, because most report a single asset. MetroPT and MetroPT2 are the exception in this collection: they cover comparable compressor subsystems and permit a genuine cross-dataset experiment [23, 24]. Section 7.4 uses that pair to separate schema compatibility from transferability, and reports the frozen and recalibrated variants separately rather than reporting a single transfer number.

The industrial datasets used by the repository are public condition-monitoring resources with different independence units and labels. MetroPT is an air-production-unit telemetry dataset for predictive maintenance [23]. MetroPT2 is a replication-style benchmark released through Zenodo [24]. SCANIA Component X is a fleet-level multivariate time-series dataset with operational histories and repair records for an anonymized component [11]. UCI Hydraulic Systems provides laboratory hydraulic test-rig cycles with component-condition targets [9]. UCI SECOM provides semiconductor process measurements with pass/fail yield labels [16]. These datasets broaden the empirical base, but they do not remove the need for estimand-specific accounting: one long compressor fault interval, one truck repair history, one hydraulic load cycle, and one wafer-level yield example are not interchangeable statistical units. The methodological gap addressed here is the absence of a reproducible layer that connects dynamical extreme diagnostics to alarm episodes, tabular condition diagnostics, sensitivity analyses, and claim constraints.

3 Method

Let $Z_t \in \mathbb{R}^p$ be the causal state vector at sample t . For compressor telemetry, Z_t contains the declared raw pressure, temperature, current, and flow channels used by the real-data diagnostic. For tabular condition-monitoring adapters, Z_t contains the declared feature vector for the cycle, vehicle, or production example.

Scaling is fitted on the training partition only. Each feature is centred by its training median and divided by a scale

$$s_j = \max\{\text{IQR}_j, \kappa (q_{0.999,j} - q_{0.001,j})\}, \quad \kappa = 0.01,$$

where all quantiles are training quantiles. The floor is not cosmetic. An interquartile range measures the spread of the central half, and a channel that idles at a fixed value between excursions has a central half made of quantisation noise: MetroPT’s `pressure_tp2` has an interquartile range of 0.004 against a 0.1-to-99.9 percentile range near ten, so its central half occupies four hundredths of one percent of its range. Dividing by that spread multiplies ordinary variation by roughly two hundred and fifty, and in a maximum-absolute- z aggregate one such channel then sets the score, its threshold and its exceedances for every other channel. Flooring the denominator at a fraction of the feature’s own range leaves well-behaved features untouched, since for them the interquartile range already exceeds κ times the range, and prevents a quantised or mostly idle channel from setting the scale for the rest. The range is measured between the 0.1st and 99.9th percentiles rather than the 1st and 99th because a channel whose excursions are rarer than one percent has a narrow 1-to-99 range as well, which would leave the floor as degenerate as the quantity it bounds.

Operating regimes R_t , when used in the sensitivity assets, are inferred without test labels. Regimes are conditioning variables for threshold estimation and are not interpreted as mechanical causes. The registered real-data matrix reported in Table 3 and Supplementary Table S5 uses a conservative train-split robust score rather than the full dangerous-region construction.

The implementation also provides dangerous-region scoring for separate diagnostics. The failure-proximal prototype variant uses only training failure intervals and records the source interval for every prototype. The unsupervised rare-state variant ignores failure labels and fits a high-distance rare-state set from training data only. These variants are available for diagnostic analysis and ablation coverage, but the registered real-data matrix in this manuscript is the conservative high-quantile score; it should not be read as evidence that both dangerous-region variants have already been optimized on every dataset.

For a target region A , the scalar observable is

$$X_t = -\log\{d(Z_t, A) + \epsilon\},$$

where d is the robust Euclidean distance in scaled coordinates and $\epsilon > 0$ prevents singularities. Larger X_t means closer proximity to the training-defined region. This is an anomaly or recurrence score, not a calibrated probability of failure.

For each regime r , a threshold $u_r(q)$ is estimated from the declared training or calibration partition at quantile q . Exceedances are $I_t(q, r) = \mathbf{1}\{X_t > u_r(q), R_t = r\}$. Runs declustering merges exceedances separated by at most m samples. The runs estimator, Ferro–Segers inter-exceedance estimator, and K-gaps-style declustering diagnostic summarize clustering under the frozen q and m settings [6, 20]. The inverse extremal index is discussed only as an asymptotic cluster-size diagnostic under the assumptions of the estimator.

Pointwise exceedances are converted into alarm episodes by the declared merge gap. A test alarm is a true event alarm only if the episode intersects the predeclared warning window for a held-out failure under the frozen matching tolerance. All event recall, event precision, false-alarm burden, duplicate-alarm count, lead time, time under warning, and utility metrics are computed after this episode conversion when event labels exist. This prevents timestamp counts from being treated as independent maintenance evidence.

What is being evaluated. Table 1 states, for each of the nineteen benchmarked methods, what its score measures, how its threshold is chosen, how exceedances become episodes, which partition its parameters are fitted on, and where extreme-value theory enters. Three facts in

that table matter for how this paper should be read, and none of them is apparent from the method names.

First, the recurrence observable $X_t = -\log\{d(Z_t, A) + \epsilon\}$ defined above is used by the three target-region methods. It is not the score of the method labelled as registered. That method scores a robust maximum absolute z , which is a multivariate anomaly magnitude, and it shares that score with seven of the baselines; what distinguishes it from them is its event policy.

Second, extreme-value theory changes the alarms a method raises in only five of the nineteen. A tail model sets the threshold for the peaks-over-threshold baseline and for SPOT, and an estimated extremal index sets the declustering run length for the Ferro–Segers policy, the K-gaps policy and the registered method. For the remaining fourteen the extremal-index estimates reported here are diagnostics computed beside the alarms; those methods would raise identical alarms if the estimates were discarded.

Third, and following from the first two, the contribution offered here is the evaluation architecture rather than a detector. What this paper supplies is a layered decomposition that locates where a detector loses its signal, a rank separation and a shift-based null that say whether a match carries information, and an event-level protocol under which simple and elaborate methods are compared on the same terms. The dynamical-EVT material motivates the recurrence observable and supplies the declustering rule, and the empirical results are evidence about what those produce under this protocol on two failures. They are not evidence that a robust anomaly score thresholded and declustered constitutes a dynamical-EVT detector, and the paper does not claim it.

Table 1: What each benchmarked method is. The score column says what the quantity measures rather than how it is named: proximity to a recurrent state, the magnitude of an anomaly, a reconstruction error, or a probability in a tail. The last column says where extreme-value theory enters. It changes the alarms a method raises only when a tail model sets the threshold or an estimated extremal index sets the declustering run length; elsewhere the extremal-index estimates reported in this paper are diagnostics computed beside the alarms, and the alarms would be identical without them.

Method	Family	Score measures	Threshold rule	Event policy	Fitted on	Role of EVT
Engineering threshold	baseline	anomaly magnitude	fixed physical $3z$	merge gap	none (fixed a priori)	diagnostic only
Best sensor threshold	baseline	anomaly magnitude	train quantile	merge gap	train	diagnostic only
Global empirical	baseline	anomaly magnitude	train quantile	merge gap	train	diagnostic only
Regime empirical	baseline	anomaly magnitude	regime conditioned quantile	merge gap	train	diagnostic only
Classical POT/GPD	baseline	anomaly magnitude	gpd return level	merge gap	train	sets threshold
Fixed-run declustering	baseline	anomaly magnitude	train quantile	fixed run declustering	train	diagnostic only
Ferro–Segers policy	baseline	anomaly magnitude	train quantile	ferro segers run length	train	sets declustering
K-gaps policy	baseline	anomaly magnitude	train quantile	k gaps run length	train	sets declustering
SPOT-style	baseline	tail probability	spot zero	merge gap	train	sets threshold
Isolation Forest	baseline	anomaly magnitude	train quantile	merge gap	train	diagnostic only
Robust changepoint	baseline	change magnitude	train quantile	merge gap	train	diagnostic only

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Method	Family	Score measures	Threshold rule	Event policy	Fitted on	Role of EVT
Linear autoencoder	baseline	reconstruction error	train quantile	merge gap	train	diagnostic only
Compact autoencoder	baseline	reconstruction error	train quantile	merge gap	train	diagnostic only
Conformal anomaly	baseline	tail probability	conformal level	merge gap	train (calibration split)	diagnostic only
Empirical horizon risk	baseline	estimated event probability	train quantile	merge gap	train	diagnostic only
Dynamical EVT score	registered	anomaly magnitude	train quantile	extremal index declustering	train	sets declustering
Failure prototype	target region	recurrence proximity	train quantile	merge gap	train	diagnostic only
Rare-state region	target region	recurrence proximity	train quantile	merge gap	train	diagnostic only
Negative control	negative control	recurrence proximity	train quantile	merge gap	train	diagnostic only

Warning-window rank separation and its null. To separate a score that carries no pre-onset ordering from a threshold that discarded the ordering a score did carry, we compute the probability that a sample drawn from the declared horizon before a failure onset outranks one drawn from the rest of the test split. This is the Mann–Whitney statistic on midranks, so it is invariant to any monotone transformation of the score and comparable across methods; 0.5 is chance. Midranks matter here because industrial telemetry ties heavily during idle operation, and breaking ties by position would make a constant score appear informative purely because the warning window sits at one end of the series.

A recall of one is only evidence of anticipation if a detector could not have achieved it by alarming often enough. We therefore compute a chance-match probability by circularly shifting the entire alarm series by a uniformly random offset and asking how often at least one episode still intersects a matching window. The shift preserves the episode count, the episode durations, and the burstiness and regime structure of their spacing, and randomises only the alignment to the failure. Because an episode $[s, s + L]$ intersects a window $[a, b]$ exactly when $(s + d) \bmod n$ falls in an interval of length $b - a + L + 1$, the quantity is a measure of a union of circular intervals and is computed exactly rather than sampled.

We do not use a homogeneous Poisson model for this null. Poisson arrivals are independent and occur at a constant rate, whereas these episodes are produced by declustering a temporally dependent score, their rate varies with operating regime, and the merge rule induces further dependence between neighbouring episodes. A null that assumes extremes do not cluster cannot calibrate a paper about clustered extremes.

Event matching. Let $P = (p_1, \dots, p_m)$ be the alarm episodes after merging and $F = (f_1, \dots, f_k)$ the labelled failures. For failure f with onset f^{start} , the matching window under horizon h , pre-tolerance τ_- and post-tolerance τ_+ is

$$W(f) = [\max\{0, f^{\text{start}} - h - \tau_-\}, f^{\text{start}} + \tau_+].$$

The window ends at onset plus τ_+ , not at the end of the failure interval, so an alarm raised well inside a multi-day failure is not counted as a warning. An alarm p is *eligible* for f when $p \cap W(f) \neq \emptyset$; an alarm that begins before the window and extends into it is eligible, since

only intersection is required. Matching is one-to-one: each alarm may be assigned to at most one failure and each failure to at most one alarm, chosen by a maximum-cardinality assignment over the eligible pairs, with ties broken toward the longer lead time. Alarm reuse across failures is available in the protocol but disabled here. Every alarm then falls in exactly one of three classes:

matched	assigned to some failure;
duplicate	eligible for some failure but not assigned;
false	eligible for no failure.

Event recall is matched failures over $|F|$. Event precision is

$$\text{precision} = \frac{\# \text{matched}}{\# \text{matched} + \# \text{duplicate} + \# \text{false}} = \frac{\# \text{matched}}{m},$$

so duplicates *do* enter the denominator and precision is matched alarms over all alarms raised. This is deliberate: an operator responds to a duplicate exactly as to any other alarm, and excluding duplicates would let a detector improve its precision by emitting more alarms near a failure it has already matched.

These conventions determine the reported numbers, and one consequence deserves emphasis. Because episodes are formed by merging before matching, a detector that is permanently on produces a single episode that is matched, with no duplicates and no false alarms, and therefore attains precision and recall of one. Episode-level metrics cannot detect that state, which is why every alarm-burden comparison in this paper reports time under warning alongside episode counts, and why the benchmark flags any method whose warning exposure exceeds half the test period.

Joint utility. The control comparisons use one scalar, declared before the comparison that consumes it. For a detector with detection indicator $D \in \{0, 1\}$, false alarms per operating day a , warning exposure fraction c , and lead time ℓ in samples,

$$U = w_D D - w_a a - w_c c + w_\ell D \frac{\min\{\ell, \ell_{\max}\}}{\ell_{\max}},$$

with $w_D = 100$, $w_a = 1$, $w_c = 10$, $w_\ell = 5$ and $\ell_{\max} = 7200$ samples. The units are therefore: one detection is worth 100, one false alarm per operating day costs 1, being under warning for the whole test period costs 10, and a lead time at or beyond two hours adds 5. The ordering is the substantive claim. Detection dominates by construction, so a method must avoid more than a hundred false alarms per day to justify missing a failure, which no method here approaches; the lead-time reward is capped and paid only on detection, so it cannot make a missed failure look competitive. Section 7 reports a sensitivity sweep varying each weight by factors from 0.25 to 4, and the component metrics are reported separately so a reader who prefers different weights can recompute.

The score-stratification target for horizon h is

$$Y_t^{(h)} = \mathbf{1}\{\text{a labelled event begins within } h \text{ after } t\}.$$

In the current evidence, model outputs are anomaly ranks or recurrence scores. Probability-calibration language is therefore restricted: Brier-style summaries are reported only as diagnostics of score stratification unless a separate calibration partition contains enough positive independent units for an explicit probability map.

The leakage contract is: training features cannot contain held-out failure samples; scaling, regimes, target regions, thresholds, lag selection, merge-gap selection, and calibration cannot use test events; SCANIA vehicles cannot appear in multiple partitions; and feature windows

cannot cross maintenance-reset boundaries. Table S9 gives the terminal audit result for each registered check. Under the current registered diagnostic the algorithm is:

1. Split entities or intervals into training, validation, calibration, and test partitions.
2. Fit scaling and robust score thresholds on allowed partitions only.
3. Select q, m , and merge gap on validation/calibration partitions only where used.
4. Freeze all parameters and score the held-out test partition causally.
5. Extract exceedance clusters, merge alarm episodes, and match them to labelled events.
6. Report event metrics, uncertainty by independent unit, and non-estimable cases.

The practical complexity for scoring is $O(npK)$, where n is the number of samples or entities, p is the state dimension, and K is the number of stored prototypes or rare-state summaries. Thresholding and runs declustering are linear in n . The streaming-compatible parts are causal feature construction, scaling with frozen parameters, scoring, thresholding, and alarm merging. Prototype construction, model selection, and final event matching are offline evaluation steps.

4 Simulation Study

The simulation layer is designed to falsify overconfident interpretations before industrial data are used. The registered systems include independent extremes, periodic dynamics, noisy periodic dynamics, regime mixtures, missingness perturbations, smoothing and downsampling perturbations, and degradation-like cyclic trajectories. Each simulation cell records its configuration, random seed, threshold grid, run-length grid, estimator status, and raw outputs sufficient to regenerate the paper assets.

4.1 Estimators under comparison

Six estimators of the extremal index are compared: runs declustering, the Ferro–Segers intervals estimator, K-gaps, the reciprocal mean block-cluster size, the disjoint-blocks estimator, and a no-declustering reference fixed at $\theta = 1$. The last is not a competitive estimator but a control: any procedure that cannot separate a clustered process from an independent one should be indistinguishable from it. The runs and K-gaps estimators are tuned by the declustering run length; the block-based estimators are tuned by the block size, which is set to $\lceil \sqrt{n} \rceil$ so that the tuning parameter is declared rather than fitted.

The reciprocal mean cluster size is computed over disjoint blocks rather than over runs-declustering clusters. This is deliberate. When clusters are taken from runs declustering they partition the exceedances exactly, so the reciprocal mean cluster size reduces algebraically to the runs estimator and carries no independent information; reporting the two as separate estimators would present one quantity twice.

4.2 Reference extremal indices

Bias, root mean squared error and coverage all require a reference value of θ . Closed-form references exist only for the independent systems and for the noiseless logistic map at its periodic target, where $\theta = 1/2$. Every noisy configuration lies outside those cases, so references there are established numerically from a long realisation of 4×10^5 samples.

The numerical construction is deliberately conservative, because a reference obtained from one declustering rule is not independent of the estimators being judged: scoring against a runs-based reference would favour the runs estimator by construction. Two structurally different long-run estimates are therefore computed, one by runs declustering at a short run length and one by disjoint blocks at a block scaled to hold roughly one expected exceedance. A numerical reference is accepted only where the two agree within 0.05; otherwise the configuration

is recorded as one where θ could not be established, and it contributes no bias, RMSE or coverage evidence. Both long-run settings are calibrated against the two systems whose value theory already fixes, recovering 0.495 against a theoretical 0.5 and 0.961 against a theoretical 1.0.

Theoretical and numerical references are never pooled within a reported row, and every table states which kind it used.

4.3 Interval estimation

None of the six estimators has a usable closed-form interval in this setting, so intervals are obtained by a circular block bootstrap over the exceedance indicator series. Resampling whole blocks preserves the short-range dependence that the extremal index measures; an independent bootstrap would destroy exactly the structure under study and return intervals centred on one. Resamples that fail to yield an admissible estimate are counted rather than discarded, so the reported resample success rate is honest.

4.4 What this grid can and cannot establish

The simulation evidence is not uniform across the clustering regimes, and the reported scope is limited accordingly. The high-replication grid runs at a single noise level, and at that noise level the numerically established references sit close to independence: of its nine referenced process families, eight have $\theta \geq 0.9$. Its 500 replications per cell therefore buy precision about *near-independent* processes. Clustered evidence does exist — the cyclic degradation family is established between $\theta = 0.049$ and $\theta = 0.175$, and the smoothing family at $\theta = 0.272$ — but the strongly clustered cells carry far fewer replications, so their coverage estimates are correspondingly imprecise.

Accordingly, this study reports the high-replication grid as evidence about near-independent processes, and treats the clustered cells as supporting rather than primary evidence. Estimator failures, non-estimable quantities and unestablished references are preserved in the artifacts as part of that scope statement rather than removed from it.

5 Datasets

The datasets are assigned roles before interpretation. MetroPT and MetroPT2 are the primary event-level case studies because they contain ordered compressor telemetry and labelled failure periods [23, 24]. They support early-warning, alarm-episode, recurrence, and per-failure questions, but their independent event count is small. SCANIA Component X is an external vehicle-level stress test with irregular fleet histories, repair labels, and censoring [11]. Hydraulic Systems is a supplementary laboratory condition-monitoring demonstration with load-cycle component states [9]. SECOM is a supplementary high-dimensional production-quality demonstration with wafer-level yield labels [16].

This role assignment prevents a common mistake: row counts do not become independent failures. Table S8 records the empirical role, sampling structure, independent unit, independent positives, rows, and target type used for interpretation. The independent unit column is the primary evidential quantity for empirical interpretation.

The supported estimands differ. MetroPT-style rows address event-level early warning on temporal telemetry splits. SCANIA addresses vehicle-level repair-risk ranking on a published test split. Hydraulic Systems addresses cycle-level component-condition detection. SECOM addresses production-example yield-failure detection. Hydraulic Systems and SECOM are therefore not interpreted as event-level replications of the compressor task. Broader predictive claims require coherent independent units and comparable targets, not only successful data preparation.

6 Experimental Protocol

The evaluation protocol is frozen before final experiment execution. It declares the train, validation, calibration, and test partitions; the exclusion and maintenance-boundary fields; warning horizons; alarm merge gap; matching tolerance; matching method; bootstrap unit; primary metrics; secondary metrics; and event utility costs.

What “declared” means here. Throughout this paper, a quantity described as declared or protocol-defined is one fixed in a versioned configuration file and frozen before the final execution of the experiments it governs. It is not preregistered in the strict sense: there is no time-stamped record deposited with an external registry before analysis began, and the reader should not read these terms as implying one. What the configuration does establish is that the reported results were not produced by searching over partitions, horizons, thresholds or metrics after seeing outcomes, and that any such search would be visible in the repository history.

Three groups of choices should be distinguished, because they carry different evidential weight. The partitions, warning horizon, merge gap, matching tolerance, matching rule and metric definitions were fixed before the experiments reported here and have not changed since. The utility weights were declared before the control comparison that uses them, and their sensitivity is reported rather than assumed. The warning-window rank separation, the chance-match null, the detection-conditioned controls, the feature-compatibility rule and the failure-layer taxonomy were added in response to earlier rounds of review, after results from previous versions of this work had been seen; they are diagnostic instruments introduced to interrogate an existing finding, not hypotheses fixed in advance, and Section 8 treats their output accordingly.

Primary event metrics are event precision, event recall, event F_1 , false-alarm events per operating day, duplicate alarms per failure, median warning lead time, and horizon Brier score. Secondary metrics include pointwise ranking metrics, time under warning, and event utility. Baselines are required to use the same causal feature availability and the same frozen event conversion, so a model is not rewarded for using future information or a more favorable matching rule.

Uncertainty is attached to the independent unit declared by the dataset and protocol. For simulation cells, the unit is the Monte Carlo repetition or registered replicate. For MetroPT-style case studies, the unit is the failure episode or operating interval, not the timestamp. For SCANIA, the relevant unit is the vehicle or repair history defined by the dataset release. For Hydraulic Systems, it is the load cycle with component-condition labels. For SECOM, it is the wafer or production example with a pass/fail yield outcome. Multiplicity is handled by keeping primary metrics and hypotheses predeclared and by reporting dataset-specific estimands.

The registered real-data matrix has three audit layers. The verification layer checks local raw and processed manifests, row counts, schema summaries, and declared independence units. The industrial-result layer writes one conservative diagnostic row per registered real dataset. The evidence-scope layer combines both outputs into a machine-readable summary of datasets, workflows, sampling structures, independent units, diagnostic statuses, and claim boundaries. This protocol is intentionally strict: a broader dataset base strengthens external coverage only when the estimand differences remain visible.

7 Results

7.1 Simulation validity and estimator stability

The first research question asks whether the extremal-index diagnostics are stable enough to interpret. The high-replication simulation artifact covers fourteen process families with 500 replications per configured cell, and a focused coverage artifact uses 1000 replications on a pre-specified subset. The six estimators of Section 4 are compared against theoretical references

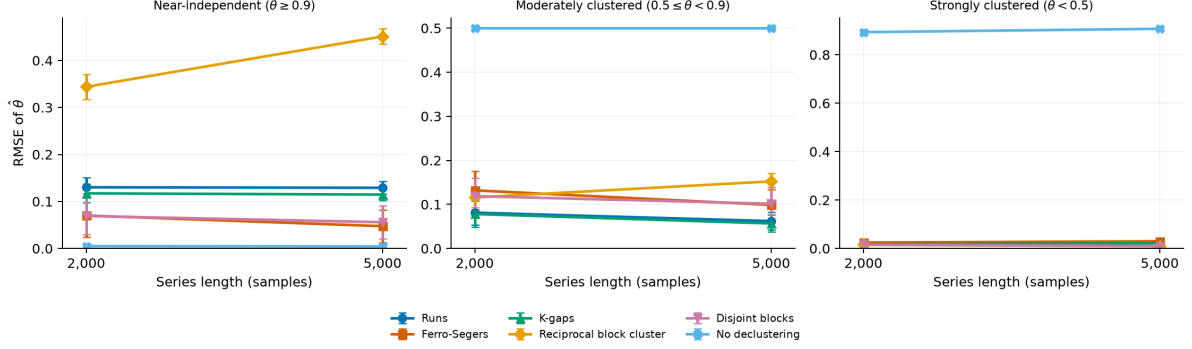


Figure 1: Finite-sample RMSE of the extremal-index estimate by series length, faceted by clustering regime. Error bars are 1.96 bootstrap standard errors. Panels use independent vertical scales because the no-declustering control has RMSE $|1 - \theta|$ by construction. Reference values are $\theta = 1$ (theoretical) for the near-independent panel, $\theta = 1/2$ (theoretical) for the moderately clustered panel, and numerically established values below 0.3 for the strongly clustered panel.

where a closed form exists and numerical references otherwise, with unestablished configurations excluded from every performance quantity. Table 2 reports one representative process per clustering regime; the full grid and the reference-availability audit are in the supplement.

The central result is that the estimator ranking *inverts* between regimes. For the independent process, where $\theta = 1$, runs and K-gaps record interval coverage of 0.40 and 0.39 against a nominal 0.95, with biases of -0.133 and -0.117 . For the periodic target, where $\theta = 1/2$, the same two estimators are the least biased of the six, at -0.010 and $+0.003$, with coverage of 0.844 and 0.850. A reader who saw only the near-independent rows would conclude against precisely the estimators that work once extremes actually cluster.

The apparent superiority of Ferro–Segers and disjoint blocks at $\theta = 1$ requires care. Both attain coverage of 1.000 there, but with boundary-clipping probabilities between 0.50 and 0.70: their intervals contain the truth largely because the estimate saturates at the admissible boundary $\theta = 1$, not because they are calibrated. Figure 3 reports the clipping share beside the non-estimability share for this reason, and coverage at $\theta = 1$ should not be read without it. The undercoverage of runs and K-gaps has the same origin: at the boundary the estimate can only err downwards, so a symmetric percentile interval misses systematically. Coverage comparisons at $\theta = 1$ are structurally distorted for every estimator, which is a further reason to treat the near-independent grid as bounded evidence.

The no-declustering control behaves exactly as intended, with zero bias at $\theta = 1$ and bias $+0.500$ and $+0.728$ in the two clustered rows, confirming that the comparison is measuring extremal clustering rather than an artifact of the harness.

Threshold and run-length sensitivity in Figure 4 shows that clustering conclusions depend on q and m ; the tabulated grid is Supplementary Table S11. That supports a negative-result interpretation: regime-conditioned dynamical-EVT diagnostics are sensitive design objects, not model-free evidence of impending failure.

7.2 Alarm conversion and failed-detection regions

The second research question asks whether declustering and alarm merging reduce duplicate alarms while preserving event recall. In the registered threshold grid, event recall was zero at all reported quantiles, with false-alarm burdens between 133.92 and 168.48 alarm episodes per operating day. This is a failed-detection region under the current observable, threshold grid, and matching rule; it shows that alarm conversion can destroy utility. Across all registered combinations of alarm merge gap and post-onset matching tolerance, event recall, precision,

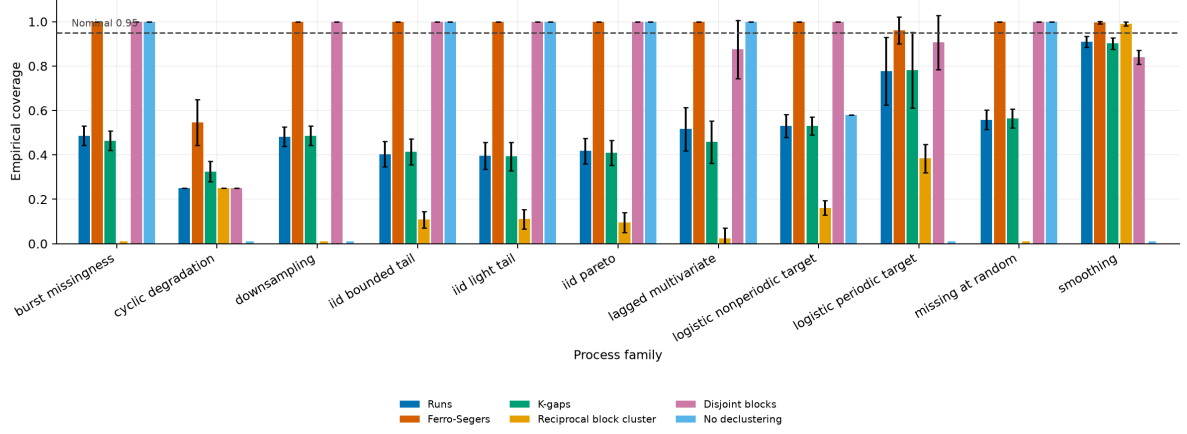


Figure 2: Empirical coverage of the bootstrap interval by process family against the nominal 0.95 level, with 1.96 binomial standard errors. Families whose reference extremal index could not be established are omitted, since coverage is undefined without a reference.

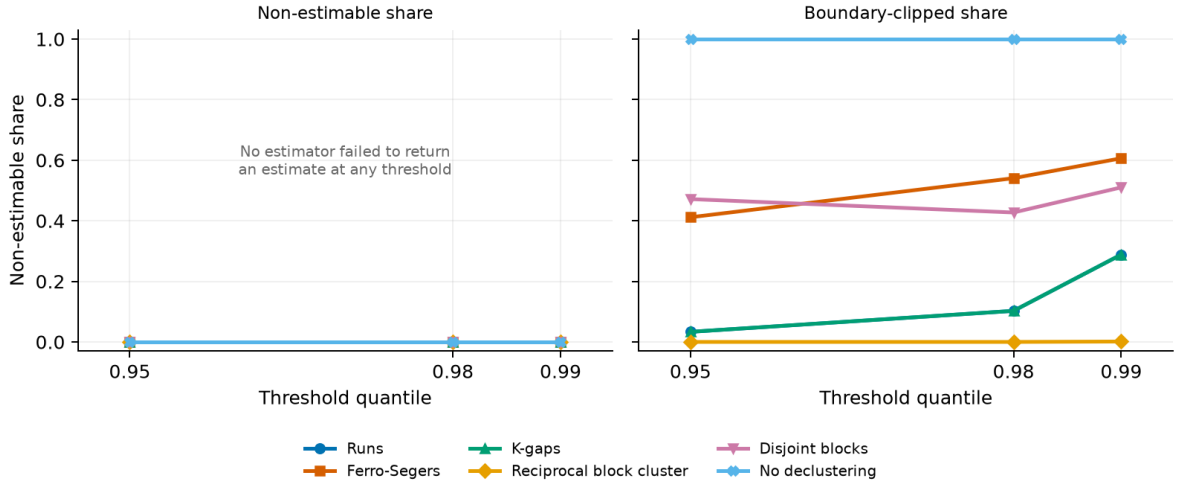


Figure 3: Non-estimability and boundary clipping against threshold quantile. No estimator failed to return a value at any threshold; the informative panel is the clipped share, which explains the coverage of the block-based estimators near $\theta = 1$.

and F_1 remained zero, and relaxing post-onset tolerance did not alter the result. Both grids are reported in the supplement rather than as main-text figures, because a zero-recall surface adds no scientific information beyond the statement that it is zero.

7.3 Complete held-out event-level benchmark

Nineteen methods were run on both compressor datasets under one shared alarm policy: identical warning horizon, matching tolerance, matching rule, operating exposure and independent unit, verified by the fairness audit in Supplementary Table S4. Alarm merging is treated as part of the detector rather than the protocol, since four methods exist precisely to vary declustering; the audit reports which event conversions were used instead of requiring one. Supplementary Table S2 reports the complete held-out event-level benchmark under the shared alarm policy, and Supplementary Table S3 records each method’s score, threshold rule, event policy, merge gap, parameter count, calibration status and tuning partition.

Five conclusions follow, and none of them is favourable to the registered method.

Table 2: Finite-sample extremal-index estimator performance for one representative process per clustering regime, aggregated over the threshold span shown. Coverage is measured against a nominal 0.95 level; the failure column is the share of replicates returning no estimate. A dagger marks a numerically established reference extremal index; unmarked references are closed-form. The full grid is in the supplement.

Estimator	n	Threshold	Bias	RMSE	Coverage	Fail.
<i>iid light tail, $\theta_{\text{ref}} = 1.000$</i>						
Runs	5000	0.95-0.99	-0.133	0.139	0.397	0.000
Ferro-Segers	5000	0.95-0.99	-0.025	0.051	1.000	0.000
K-gaps	5000	0.95-0.99	-0.117	0.123	0.393	0.000
Reciprocal block cluster	5000	0.95-0.99	-0.423	0.427	0.111	0.000
Disjoint blocks	5000	0.95-0.99	-0.030	0.058	1.000	0.000
No declustering	5000	0.95-0.99	0.000	0.000	1.000	0.000
<i>logistic periodic target, $\theta_{\text{ref}} = 0.500$</i>						
Runs	5000	0.95-0.99	-0.010	0.072	0.844	0.000
Ferro-Segers	5000	0.95-0.99	0.064	0.115	0.933	0.000
K-gaps	5000	0.95-0.99	0.003	0.067	0.850	0.000
Reciprocal block cluster	5000	0.95-0.99	-0.113	0.134	0.578	0.000
Disjoint blocks	5000	0.95-0.99	0.065	0.110	0.839	0.000
No declustering	5000	0.95-0.99	0.500	0.500	0.000	0.000
<i>smoothing, $\theta_{\text{ref}} = 0.272^\dagger$</i>						
Runs	2000	0.98	-0.012	0.036	0.910	0.000
Ferro-Segers	2000	0.98	0.019	0.076	0.996	0.000
K-gaps	2000	0.98	-0.009	0.039	0.902	0.000
Reciprocal block cluster	2000	0.98	-0.015	0.041	0.990	0.000
Disjoint blocks	2000	0.98	0.021	0.053	0.840	0.000
No declustering	2000	0.98	0.728	0.728	0.000	0.000

Table 3: Compact held-out event-level benchmark: the best method in each category per dataset, selected on event F_1 with alarm burden as the declared tie-break. A method flagged as effectively always on cannot win a category, because episode-level precision and recall are degenerate for a permanently-on detector. The complete benchmark over all nineteen methods, including any excluded on that ground, is in Supplementary Table S2.

Category	Method	Det.	Precision	FA/day	Lead	Utility
<i>MetroPT</i>						
Simple	Single-sensor threshold	yes	0.000441	90.9	2,333	-452
Classical EVT	Classical POT/GPD	yes	0.000395	101	3,334	-505
Anomaly	Conformal anomaly	yes	0.000394	102	3,343	-506
Learned	Linear autoencoder	yes	0.000212	189	3,433	-942
Registered	Recurrence score	yes	0.000361	111	3,343	-552
Prototype region	Failure-prototype region	no	0	0		-2
Rare-state region	Rare-state region	yes	0.00112	35.6	3,414	-177
<i>MetroPT2</i>						
Simple	Single-sensor threshold	yes	0.00282	21.2	3,444	-69.3
Classical EVT	Fixed-run declustering	yes	0.00113	53.3	3,529	-175
Anomaly	Conformal anomaly	yes	0.00113	53.3	3,529	-175
Learned	Nonlinear autoencoder	yes	0.00128	47.3	3,452	-155
Registered	Recurrence score	yes	0.00112	54	3,529	-177
Prototype region	Failure-prototype region	yes	0.333	0.121	7,264	0.6
Rare-state region	Rare-state region	yes	0.00145	41.8	7,264	-137

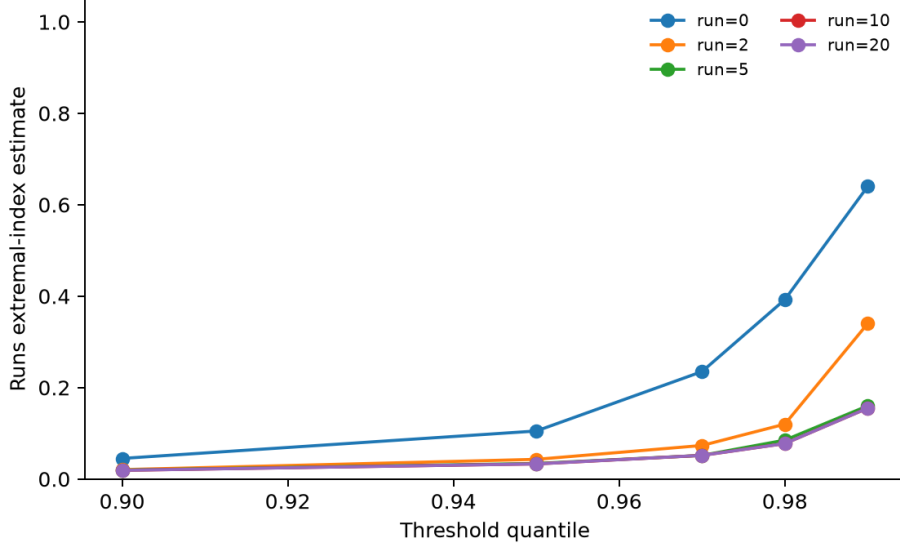


Figure 4: Threshold and run-length stability diagnostic for the registered recurrence observable.

No method is operationally plausible on either dataset. Every method that detects the held-out failure does so with event precision below 0.7%, and most below 0.05%. The best event utility achieved by any detecting method is negative on both datasets. A maintenance function receiving these alarms would be responding overwhelmingly to false positives.

The best simple baseline wins on both datasets. On MetroPT the best simple policy is the single-sensor threshold, at 90.9 false alarms per operating day against the registered score’s 105.3; the fixed engineering threshold reaches a lower false-alarm rate of 62.0 but is flagged as effectively always on, holding the system under warning for 75.1% of the test period, so it is excluded from selection. On MetroPT2 the single-sensor threshold is best again and by a wide margin, at 21.2 false alarms per day against 51.5 for the registered score, with more than double its event precision.

Classical EVT and anomaly baselines do not change the picture. The generalised Pareto return level, the three declustering policies, SPOT, the isolation forest, the change-point detector, both autoencoders and the conformal detector all sit within the same precision decade as the registered score. Making the tail model or the declustering rule more sophisticated does not recover operational value that the alarm conversion has already destroyed.

The registered recurrence score adds no measurable event-level value. It is beaten by a simple single-sensor threshold on both datasets, on precision, on false alarms per day and on event utility. Its distinguishing feature is not accuracy but exposure: at 0.12% and 0.06% of the test period under warning it is far less intrusive than the engineering threshold, which is a real property but not the early-warning claim the method was proposed to support.

The one exception is the failure-prototype region on MetroPT2. There it attains the best precision of any method, 0.0064, at 9.4 false alarms per day and the best utility, -30.1 . Under this seven-feature construction the same region raises no alarms at all on MetroPT and misses the failure entirely. Section 7.4 shows that the MetroPT non-detection is a property of the feature set rather than of the method, and that the region does transfer in one direction.

Alarm burden must be read on two axes that can disagree. Episode counts measure how often an operator is interrupted; time under warning measures how long the system is held in an alarmed state. On MetroPT the engineering threshold raises fewer episodes than the registered score (1551 against 2631) while holding the system under warning for 75.1% of the test period against the registered score’s 0.12%. A detector that merges its exceedances into few long alarms therefore looks better on episode counts and far worse on exposure. Because

episode-level precision and recall are computed after merging, a permanently-on detector attains a recall and precision of one with no unmatched episodes to count as false alarms; exposure is the only reported quantity that exposes it. Every alarm-burden comparison in this paper is therefore reported with both axes, and the supplementary benchmark flags any method whose warning exposure exceeds half the test period.

7.4 Cross-dataset target-region transfer

Transfer is tested by fitting a target region on one compressor dataset and applying it to the other, not by comparing the two datasets’ own metrics. Four protocols are separated: *direct*, with scaling, prototypes, threshold and distance metric all frozen from the source; *threshold recalibration* on the destination validation split; *scaling recalibration* on the destination training split with the source prototypes retained; and a *destination refit*, which is the within-dataset reference that degradation is measured against. Table 4 reports all four in both directions.

Compatibility is checked before any region crosses. The two datasets share an identical schema and a compatible sampling cadence, 1.000 against 0.991 seconds, and five of the seven candidate features are admissible. Two are not: `reservoir_pressure` and `flowmeter` have effectively zero overlap between the datasets, with standardised median shifts of 122 and 503 interquartile ranges. Those two channels occupy disjoint ranges across the datasets and are excluded rather than silently mapped, so the transfer runs on the five compatible features.

Transfer is strongly asymmetric. Applying the MetroPT region directly to MetroPT2 reaches event precision 0.0057 at 10.5 false alarms per operating day, which is better than a region fitted on MetroPT2 itself (0.0022 at 27.0); the degradation against the within-dataset refit is negative. The reverse direction fails: the MetroPT2 region applied directly to MetroPT occupies 50.3% of the test period at precision 0.00033, an alarm that is on more often than off. Recalibration does not act uniformly. It rescues the failing direction, lifting MetroPT2 to MetroPT from 0.00033 to 0.00108, and it degrades the working one, dropping MetroPT to MetroPT2 from 0.0057 to 0.0012.

Figure 5 shows why the two directions differ. In the working direction the destination’s failure states lie closer to the transferred prototypes than its normal operation, which is the ordering a usable region requires. In the failing direction that ordering inverts: MetroPT’s normal operation sits closer to the MetroPT2 prototypes than MetroPT’s own failure does, so the frozen threshold admits ordinary operation first and the region is occupied half the time. The failure is therefore not a matter of threshold placement, and no recalibration of the threshold alone can repair it.

The feature restriction also explains the MetroPT non-detection reported above. Holding the threshold rule fixed and varying only the feature set, the five compatible features detect the held-out MetroPT failure with 815 alarm episodes and recall 1.0, while all seven produce zero alarms and zero region occupancy. The seven-feature run reproduces the benchmark exactly, so the two constructions agree and the difference is attributable to the feature set alone. Including the two channels with no cross-dataset distributional support suppresses detection entirely on MetroPT.

The consequence for the paper’s central heterogeneity claim is a narrowing rather than a reversal. A recurrence region does carry information across these two compressors in one direction, and better than refitting locally. But the direction matters, recalibration can hurt as easily as help, and part of what previously looked like dataset heterogeneity was a feature-selection artifact. The two datasets still cannot be pooled as equivalent evidence, and one directional success across two failures is not a transferability result.

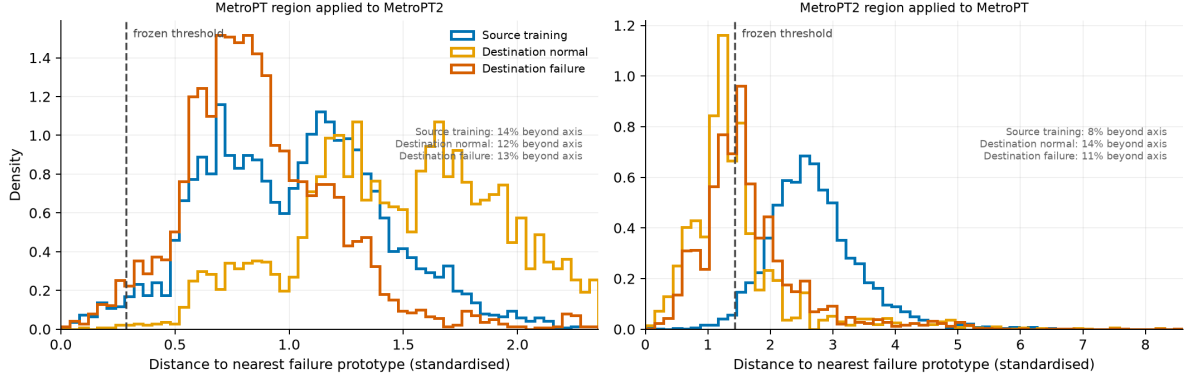


Figure 5: Distance to the nearest transferred failure prototype, for the source’s training rows, the destination’s normal operation, and the destination’s failure states, against the frozen threshold. The horizontal axis is truncated at six times the threshold because the distance distribution is heavy-tailed; the share of each population beyond the axis is annotated. A transferable region requires destination failure states to sit closer than destination normal operation, which holds in the left panel and is inverted in the right.

Table 4: Cross-dataset target-region transfer. Each row applies a region fitted on the source to the destination’s held-out test split; the scaling and threshold columns state what was frozen and what was refitted. Degradation is event precision relative to the destination refit, so a negative value means the transferred region outperformed a region fitted on the destination itself. Transfer runs on the transferable subset (5 of 7). Excluded as incompatible: reservoir pressure, flowmeter.

Source	Dest.	Protocol	Scaling	Threshold	Occup.	Recall	Precision	FA/day	Degrad.
MetroPT	MetroPT	Direct	metropt	metropt	0.00513	1	0.00571	10.5	-
MetroPT	MetroPT	Threshold recal.	metropt	metropt2 validation	0.0391	1	0.00115	52.4	0.00348
MetroPT	MetroPT	Scaling recal.	metropt2 train	metropt2 train	0.0295	1	0.00172	35.1	0.000511
MetroPT	MetroPT	Destination refit	metropt2 train	metropt2 train	0.0693	1	0.00223	27	0
MetroPT	MetroPT	Direct	metropt2	metropt2	0.503	1	0.00033	121	0.000897
MetroPT	MetroPT	Threshold recal.	metropt2	metropt validation	0.0117	1	0.00108	37.1	0.000147
MetroPT	MetroPT	Scaling recal.	metropt train	metropt train	0.0144	1	0.00102	39.3	0.000208
MetroPT	MetroPT	Destination refit	metropt train	metropt train	0.0198	1	0.00123	32.6	0

7.5 Matched negative controls

Table S7 compares the observed failure-prototype alarm burden with 500 deterministic matched-control draws for each control family and dataset. These controls preserve occupancy, exposure length, temporal shifting, rare-region structure, or block structure while breaking the specific relationship to the held-out failure. The comparison is descriptive because only two independent held-out compressor failures are available.

Burden alone is not a sufficient basis for that comparison. A region that raises no alarms

attains a false-alarm rate of zero while missing the failure, so comparing burden without conditioning on detection rewards a detector for not detecting. Table 5 therefore reports the control population’s detection rate on every row and repeats the comparison restricted to controls that detect, that match the observed recall, and that match the observed occupancy. Utility is the predeclared joint utility defined in Section 7.3: detection dominates, false alarms are charged per operating day, warning exposure is charged separately, and a capped early-warning reward is paid only when the failure was detected. Component metrics remain reported separately in Table S7.

Table 5: Detection-aware matched controls. Every comparison reports the share of control draws that detected the failure, because a burden advantage means nothing if the observed region did not detect while the controls did. Utility is the predeclared joint utility, which charges false alarms per operating day and warning exposure separately and pays a capped early-warning reward only when the failure was detected.

Data	Control family	Condition	Draws	Ctrl det. rate	Obs. de-tected	Obs. utility	Ctrl median	Percentile
MetroPT	episode label permutation	All draws	500	0	no	0	0	1
MetroPT	episode label permutation	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	episode label permutation	Recall matched	500	0	no	0	0	1
MetroPT	episode label permutation	Occupancy matched	500	0	no	0	0	1
MetroPT	phase randomised score	All draws	500	0	no	0	0	1
MetroPT	phase randomised score	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	phase randomised score	Recall matched	500	0	no	0	0	1
MetroPT	phase randomised score	Occupancy matched	500	0	no	0	0	1
MetroPT	prototype source permutation	All draws	500	0	no	0	0	1
MetroPT	prototype source permutation	Detecting only	0	0	no	0	Not estimable	Not estimable
MetroPT	prototype source permutation	Recall matched	500	0	no	0	0	1
MetroPT	prototype source permutation	Occupancy matched	500	0	no	0	0	1
MetroPT	random occupancy	All draws	500	0	no	0	0	1

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Data	Control family	Condition	Draws	Ctrl det. rate	Obs. de- tected	Obs. utility	Ctrl median	Percentile
MetroPT	random occupancy	Detecting only	0	0	no	0	Not es- timable	Not es- timable
MetroPT	random occupancy	Recall matched	500	0	no	0	0	1
MetroPT	random occupancy	Occupancy matched	500	0	no	0	0	1
MetroPT	regime matched rare region	All draws	500	0	no	0	0	1
MetroPT	regime matched rare region	Detecting only	0	0	no	0	Not es- timable	Not es- timable
MetroPT	regime matched rare region	Recall matched	500	0	no	0	0	1
MetroPT	regime matched rare region	Occupancy matched	500	0	no	0	0	1
MetroPT	time shifted prototype window	All draws	500	0	no	0	0	1
MetroPT	time shifted prototype window	Detecting only	0	0	no	0	Not es- timable	Not es- timable
MetroPT	time shifted prototype window	Recall matched	500	0	no	0	0	1
MetroPT	time shifted prototype window	Occupancy matched	500	0	no	0	0	1
MetroPT2	episode label per- mutation	All draws	500	0.17	yes	103	-1.82	1
MetroPT2	episode label per- mutation	Detecting only	85	0.17	yes	103	103	1
MetroPT2	episode label per- mutation	Recall matched	85	0.17	yes	103	103	1
MetroPT2	episode label per- mutation	Occupancy matched	500	0.17	yes	103	-1.82	1
MetroPT2	phase randomised score	All draws	500	1	yes	103	58.5	1
MetroPT2	phase randomised score	Detecting only	500	1	yes	103	58.5	1
MetroPT2	phase randomised score	Recall matched	500	1	yes	103	58.5	1

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Data	Control family	Condition	Draws	Ctrl det. rate	Obs. detected	Obs. utility	Ctrl median	Percentile
MetroPT2	phase randomised score	Occupancy matched	500	1	yes	103	58.5	1
MetroPT2	prototype source permutation	All draws	500	0.068	yes	103	-1.82	0.986
MetroPT2	prototype source permutation	Detecting only	34	0.068	yes	103	103	0.794
MetroPT2	prototype source permutation	Recall matched	34	0.068	yes	103	103	0.794
MetroPT2	prototype source permutation	Occupancy matched	429	0.068	yes	103	-1.82	0.984
MetroPT2	random occupancy	All draws	500	0.054	yes	103	-1.82	0.986
MetroPT2	random occupancy	Detecting only	27	0.054	yes	103	103	0.741
MetroPT2	random occupancy	Recall matched	27	0.054	yes	103	103	0.741
MetroPT2	random occupancy	Occupancy matched	449	0.054	yes	103	-1.82	0.984
MetroPT2	regime matched rare region	All draws	500	0.872	yes	103	65.1	1
MetroPT2	regime matched rare region	Detecting only	436	0.872	yes	103	65.3	1
MetroPT2	regime matched rare region	Recall matched	436	0.872	yes	103	65.3	1
MetroPT2	regime matched rare region	Occupancy matched	452	0.872	yes	103	65.2	1
MetroPT2	time shifted prototype window	All draws	500	0.174	yes	103	-1.82	1
MetroPT2	time shifted prototype window	Detecting only	87	0.174	yes	103	103	1
MetroPT2	time shifted prototype window	Recall matched	87	0.174	yes	103	103	1
MetroPT2	time shifted prototype window	Occupancy matched	500	0.174	yes	103	-1.82	1

The conditioning changes both results. On MetroPT the observed region missed the failure, and so did all 3000 draws across the six control families: the detection rate is zero throughout and no draw satisfies the detecting condition. Its zero alarm burden is therefore matched by every control and accompanied by no detection, which is not evidence of a favourable region.

On MetroPT2 the effect is uniform in direction but not in size. Restricting to controls

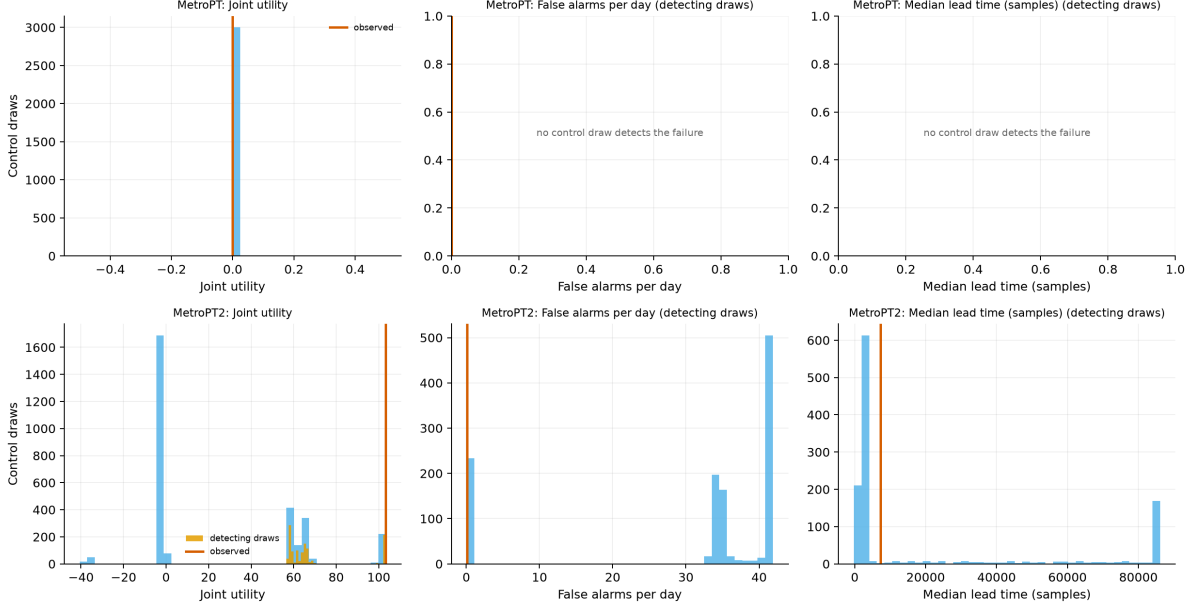


Figure 6: Matched-control distributions with the observed failure-prototype region marked. The utility panel overlays the subset of controls that detect the failure, because the comparison that matters is against those; the burden and lead-time panels are restricted to detecting draws for the same reason. On MetroPT no control draw detects, so those panels are empty by construction.

that also detect lowers the observed percentile in every family, and in two of the six it moves the observed region below the control median: from 0.654 to 0.279 against occupancy-matched random regions, and from 0.656 to 0.265 against prototype-source permutations. Against phase-randomised scores and regime-matched rare regions the observed region still exceeds every detecting draw. The families that reverse are those whose draws preserve the region’s own structure while breaking its relation to the failure, which is the comparison most informative about whether the learned region is failure-proximal; the families it still beats are those whose detection rate is near unity, where detecting the failure carries little information. Detection is not a discriminating achievement on this dataset: 92% of regime-matched rare regions and all phase-randomised scores detect the same failure.

The sensitivity sweep varies each declared weight by factors from 0.25 to 4. Pooled across families the observed percentile among detecting draws moves between 0.68 and 0.83, so the overall standing is not an artifact of the chosen weighting; nor, for the two reversing families, is the reversal.

7.6 Score-threshold-cluster-alarm decomposition

Table S6 follows every method’s evidence through the stages that separate a score from a matched alarm: test observations, threshold exceedances, extreme clusters, alarm episodes after merging, and alarm episodes that match a labelled failure. Two properties of the table govern how it should be read. The first is that the columns are not comparable across stages: observations and exceedances count samples, clusters count extreme clusters, and the remaining columns count alarm episodes, so the decline along a row is a change of unit as much as a reduction in number. The second is that the labelled failures are ground truth, not a stage the detector passes through. They enter only at the matching step, which is why they are reported separately rather than as the terminus of the flow.

Figure 7 draws that structure for four representative rows: the detector track as connected

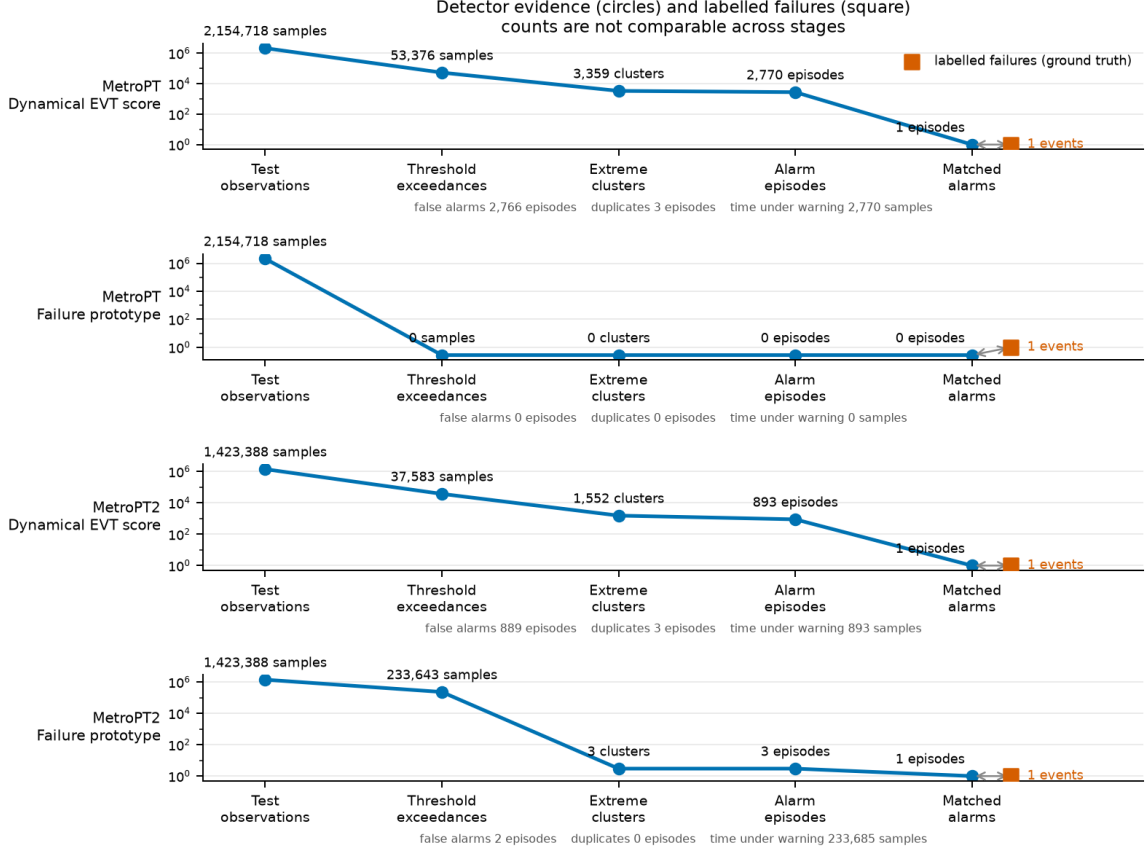


Figure 7: Score-threshold-cluster-alarm decomposition for four representative held-out event methods. Circles are the detector evidence track and the square is the labelled ground truth, which is connected to the detector only at the matching step. Each stage carries its own unit, so the counts are not comparable along a panel.

circles, the labelled failures as a separate square joined to the matched-alarm stage by the matching step alone.

The decomposition shows that the failure layer is method-specific rather than a single property of the datasets. Table 6 classifies each dataset-method pair by the earliest stage at which its evidence broke down, with the evidence, the competing explanation the evidence does not exclude, and a confidence level.

Three rows never produce an exceedance at all. The failure-prototype region on MetroPT and the empirical horizon risk on both datasets return zero exceedances, zero clusters, and zero episodes: no evidence survives the score and threshold layers, so no alarm policy could have recovered the failure. This is the layer at which the MetroPT non-detection reported above actually occurs.

Zero exceedances is consistent with two different faults, and the counts alone cannot separate them. Either the score does not rank failure-proximal states above normal ones, or it does and the threshold discarded that ranking. We separate them with the warning-window rank separation: the probability that a sample drawn from the declared horizon before onset outranks a sample drawn from the rest of the split, where 0.5 is chance. The distinction matters because the repairs differ: a score failure needs a different observable, a threshold failure needs recalibration on a score that already carries the signal.

Applied to every row, the separation shows that most methods do rank the pre-onset window above chance on both datasets. On MetroPT the registered recurrence score separates at 0.78

and on MetroPT2 at 0.86; seven of the nineteen MetroPT methods and two of the nineteen MetroPT2 methods fall below the 0.55 floor. The score layer is therefore not where most of these methods break.

This corrects an earlier version of this analysis, and the correction is worth stating because it illustrates what the layered view is for. Under a previous scaling of the observable the MetroPT separation was 0.45, below chance, and the result was reported as a score-layer failure. That scaling divided each feature by its training interquartile range with no lower bound. MetroPT’s `pressure_tp2` idles near zero, so its interquartile range is 0.004 against a range of about ten, and dividing by it multiplied ordinary variation by roughly two hundred and fifty. One channel then dominated the maximum, the 98th-percentile threshold rose above two thousand, and the ordering the other channels carried was discarded before any threshold was applied. Section 3 now floors the denominator at a fraction of each feature’s own range. The layer diagnosis is only as good as the score construction it examines, and a separation below chance is as consistent with a preprocessing fault as with an uninformative observable.

With the corrected observable the failures concentrate after alarm conversion. Of the 38 dataset-method rows, 25 match the failure and fail on burden, nine fail at the score layer, two at event matching, one at the threshold, and one is recorded as insufficient independent evidence. The registered recurrence score reaches the failure on both datasets and raises 2,770 and 893 alarm episodes to do it, at event precision 0.00036 and 0.0011.

That burden makes recall almost uninformative on its own. The circular-shift null in Section 3 preserves each detector’s episode count, episode durations and the burstiness of their spacing, and randomises only its alignment to the failure. Under it the registered score matches with probability 0.9997 on MetroPT and 0.9994 on MetroPT2. A detector alarming this often is essentially certain to overlap a one-hour window wherever that window falls, so its recall of 1.0 is not evidence that it anticipated anything. The same null separates the sparse methods sharply: the SPOT-style baseline sits at 0.034 and 0.040, and the failure-prototype region on MetroPT2 at 0.17.

One row is unlike the rest. On MetroPT2 the failure-prototype region raises three alarm episodes across 16.5 operating days, matches the held-out failure, and reaches event precision 0.333 at 0.12 false alarms per operating day. It is the only method in the benchmark with positive event utility. Three considerations keep this suggestive rather than established. Its chance-match probability is 0.17, so roughly one in six detectors with this episode structure would match by alignment alone. Its three episodes hold the system under warning for 16.4% of the test period, which is the exposure asymmetry described above working against it: few long episodes flatter an episode-level precision that an operator would not recognise. And it rests on one held-out failure. Under the same construction on MetroPT the region raises no alarms at all.

The negative-control region is the check on this procedure. It separates at 0.51 on MetroPT and 0.10 on MetroPT2, matches nothing on either, and has chance-match probabilities of 0.16 and 0.71; a control built to carry no failure information is scored as carrying none.

7.7 Method-specific failure layers

Table 6 states the failure layer for each dataset and method, together with the evidence, the competing explanation that evidence does not exclude, and a confidence level. It covers the registered recurrence score, the engineering threshold, the classical peaks-over-threshold fit, Isolation Forest, both target regions, and the best-performing baseline outside those families, selected on event utility from the benchmark rather than named in advance.

Two properties of this table are the point of it. The first is that a dataset does not have a failure layer; a dataset-method pair does. On MetroPT the failure-prototype region and the registered recurrence score fail at opposite ends of the pipeline, the first before any alarm exists and the second only after alarm conversion, and a single explanation attached to MetroPT

would be wrong for one of them. The second is that confidence varies with what the evidence can support. A zero-exceedance verdict is structural and rests on nothing but counts; a burden verdict is medium confidence because a different threshold or alarm policy is not excluded; and a method that matches with acceptable precision is recorded as insufficient independent evidence at low confidence, because one held-out failure cannot distinguish a working detector from a fortunate one.

The competing-explanation column is deliberately populated for every row. None of these verdicts is a causal proof, and for the non-event datasets the entry states explicitly that an estimand mismatch is not evidence that the method fails on those units.

Table 6: Failure layer per dataset and method. The layer is the earliest stage at which that method’s evidence broke down, taken from the score, threshold, cluster and alarm decomposition; it is not a dataset-level property, and methods on the same dataset fail at different stages. Score and threshold failures are separated by the warning-window rank separation, which is the probability that a sample in the horizon before onset outranks one outside it; 0.5 is chance. The competing explanation is what the evidence does not exclude.

Dataset	Method	Primary failure layer	Supporting evidence	Alternative not excluded	Confidence
MetroPT	Dynamical EVT score	no operationally acceptable result	separation 0.78; matched, but 2,766 false episodes give precision 0.0004, chance match 1.00; 110.9 FA/day; 0.13% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT	Engineering threshold	no operationally acceptable result	separation 0.78; matched, but 2,579 false episodes give precision 0.0004, chance match 1.00; 103.4 FA/day; 14.88% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT	Classical POT/GPD	no operationally acceptable result	separation 0.78; matched, but 2,529 false episodes give precision 0.0004, chance match 1.00; 101.4 FA/day; 1.27% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT	Isolation Forest	score failure	separation 0.48 (below chance); match among 2,577 episodes incidental, chance match 1.00; 103.2 FA/day; 14.35% exposure	a different lead time may separate; the horizon may be mis-specified	high
MetroPT	Failure prototype	score failure	separation 0.53 (at chance); no episodes raised; 0.0 FA/day; 0.00% exposure	a different lead time may separate; the horizon may be mis-specified	medium

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Dataset	Method	Primary failure layer	Supporting evidence	Alternative not excluded	Confidence
MetroPT	Rare-state region	score failure	separation 0.40 (below chance); match among 892 episodes incidental, chance match 0.62; 35.6 FA/day; 0.88% exposure	a different lead time may separate; the horizon may be mis-specified	high
MetroPT	Best sensor threshold (best baseline)	score failure	separation 0.54 (at chance); match among 2,268 episodes incidental, chance match 0.99; 90.9 FA/day; 1.16% exposure	a different lead time may separate; the horizon may be mis-specified	medium
MetroPT2	Dynamical EVT score	no operationally acceptable result	separation 0.86; matched, but 889 false episodes give precision 0.0011, chance match 1.00; 54.0 FA/day; 0.06% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT2	Engineering threshold	no operationally acceptable result	separation 0.86; matched, but 613 false episodes give precision 0.0016, chance match 1.00; 37.2 FA/day; 21.26% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT2	Classical POT/GPD	no operationally acceptable result	separation 0.86; matched, but 885 false episodes give precision 0.0011, chance match 1.00; 53.7 FA/day; 2.15% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT2	Isolation Forest	no operationally acceptable result	separation 0.90; matched, but 1,734 false episodes give precision 0.0006, chance match 1.00; 105.3 FA/day; 6.71% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT2	Failure prototype	insufficient independent evidence	matched at precision 0.333, on 1 held-out failure; 0.1 FA/day; 16.42% exposure	the result may not survive on failures not represented here	low

continued on next page

Dataset	Method	Primary failure layer	Supporting evidence	Alternative not excluded	Confidence
MetroPT2	Rare-state region	no operationally acceptable result	separation 0.89; matched, but 689 false episodes give precision 0.0014, chance match 1.00; 41.8 FA/day; 17.69% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
MetroPT2	Best sensor threshold (best baseline)	no operationally acceptable result	separation 0.75; matched, but 350 false episodes give precision 0.0028, chance match 0.19; 21.2 FA/day; 7.38% exposure	a stricter threshold or alarm policy may cut burden without losing it	medium
Hydraulic	high-quantile diagnostic	estimand mismatch	laboratory cycle states are not field failure-event onsets; 431 positives, F1 0	a redesigned target and evaluation on these units might succeed; this is not evidence that the method fails on them	high
SCANIA	high-quantile diagnostic	estimand mismatch	vehicle-level repair risk is not a compressor failure-episode onset; 5,045 vehicles, F1 0	a redesigned target and evaluation on these units might succeed; this is not evidence that the method fails on them	high
SECOM	high-quantile diagnostic	estimand mismatch	wafer yield labels are quality outcomes, not maintenance repair events; 17 positives, F1 0	a redesigned target and evaluation on these units might succeed; this is not evidence that the method fails on them	high

7.8 Timeline reconciliation

Figures 8 and 9 show each held-out failure at two scales. The upper panels cover the full test split, so every alarm episode the method raised is visible at once; the lower panels expand the failure window itself. Both scales are drawn from the same per-sample alarm series, and the shaded band in the upper panels marks the window the lower panels expand.

The two scales are shown together because either alone misleads. The local window around the MetroPT failure contains 23 alarm episodes, one of which matches the failure with a lead time of 0.93 hours. Read on its own that window describes a usable detector. Over the full

test split the same configuration raises 2,631 episodes across 24.9 operating days, of which 2,627 are false, which is 105.3 false alarms per operating day. MetroPT2 behaves the same way: 19 episodes in the plotted window against 852 over 16.5 days, 848 of them false, at 51.5 false alarms per operating day. The local view is a correct picture of the moment of detection and a misleading picture of the method.

The local windows also show that proximity is not usefulness. Of the 23 MetroPT episodes inside the plotted window, 19 fall outside the warning horizon and count as false alarms, and only one is the matched alarm; the remaining three are duplicates inside the same window. An alarm raised during the failure interval is not an early warning, so the matching window is the declared horizon before onset rather than the interval itself.

The counts in both figures are reconciled against the event-level tables rather than recomputed for the figure: the binned episode counts sum to the global total, the local window is a strict subset of the global series, the plotted threshold is the frozen test threshold, and the false-alarm rate matches the rate in Table 3. These checks are enforced in the test suite, so a figure that drifts from the tables fails the build rather than reaching a reader.

7.9 External estimands

SCANIA Component X supports a limitation the compressor datasets cannot express. Its independent unit is a vehicle and its label is a component repair, so the quantity a detector would estimate there is fleet-level repair risk rather than the onset of one machine’s failure episode. Under the conservative high-quantile diagnostic it attains $F_1 = 0$, and that number is uninformative about the method: no alarm policy on vehicle-level repair risk answers the question the held-out compressor events answer. Table 6 therefore records an estimand mismatch, with the competing explanation stating that this is not evidence the method fails on those units.

Hydraulic Systems and SECOM carry the same mismatch for laboratory cycle states and wafer yield labels respectively. All three are reported in Supplementary Section 4 rather than the main text, because a stress test whose estimand differs from the primary one cannot strengthen or weaken the primary claim.

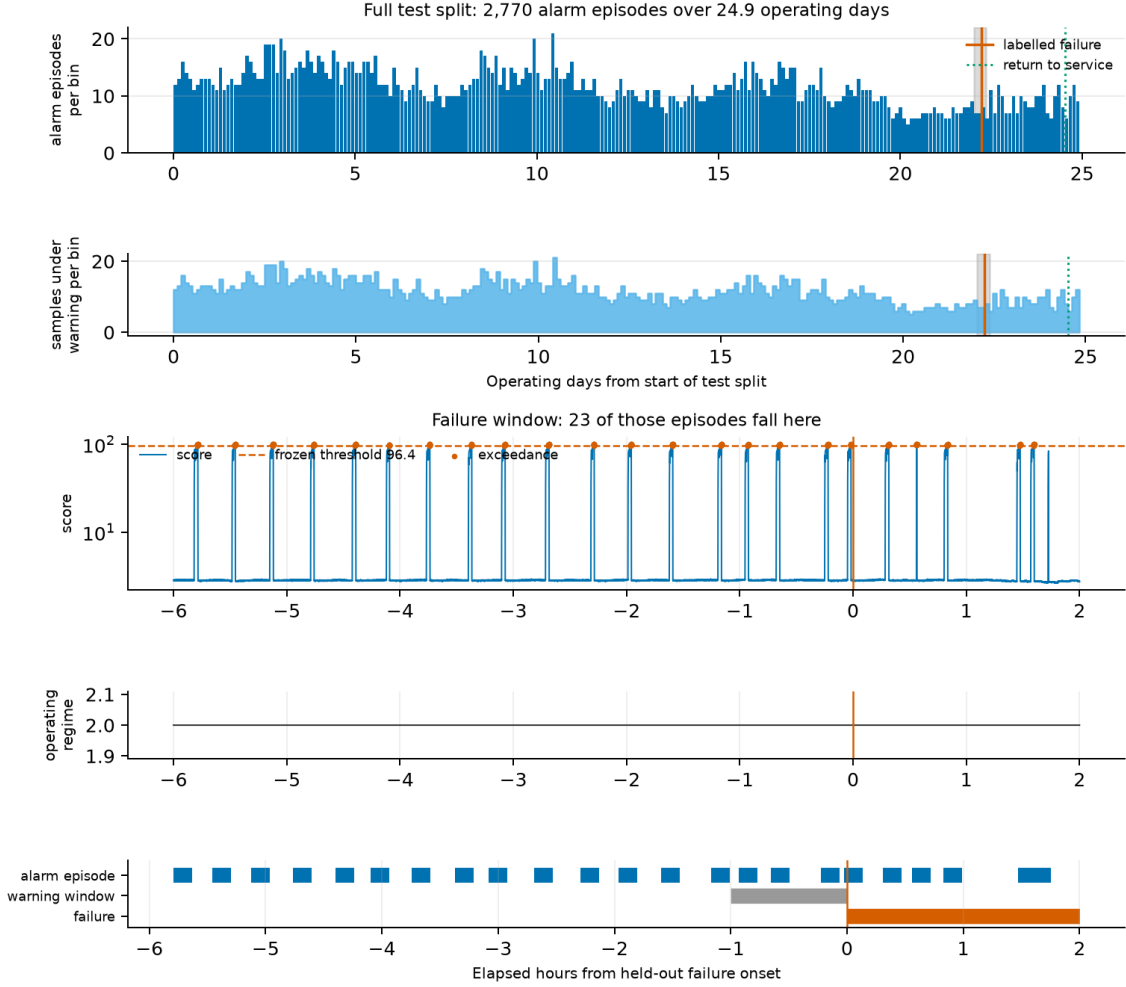


Figure 8: Global and local views of the MetroPT held-out failure `metropt_test_failure_001` under the Dynamical EVT score method (robust_max_abs_z score). Threshold quantile 0.980 gives a frozen test threshold of 96.4; declustering run length 6 samples, merge gap 0 samples, warning horizon 1.0 h. The test split runs 24.9 operating days (2,154,718 samples). Globally the method raises 2,770 alarm episodes, of which 2,766 are false, at 110.91 false alarms per operating day; 23 episodes fall inside the plotted failure window and 19 of those are false. The matched alarm leads the failure by 0.93 h. Time under warning is 2,770 samples (0.13% of the split). The only maintenance boundary the processed data records is the end of the single labelled failure interval.

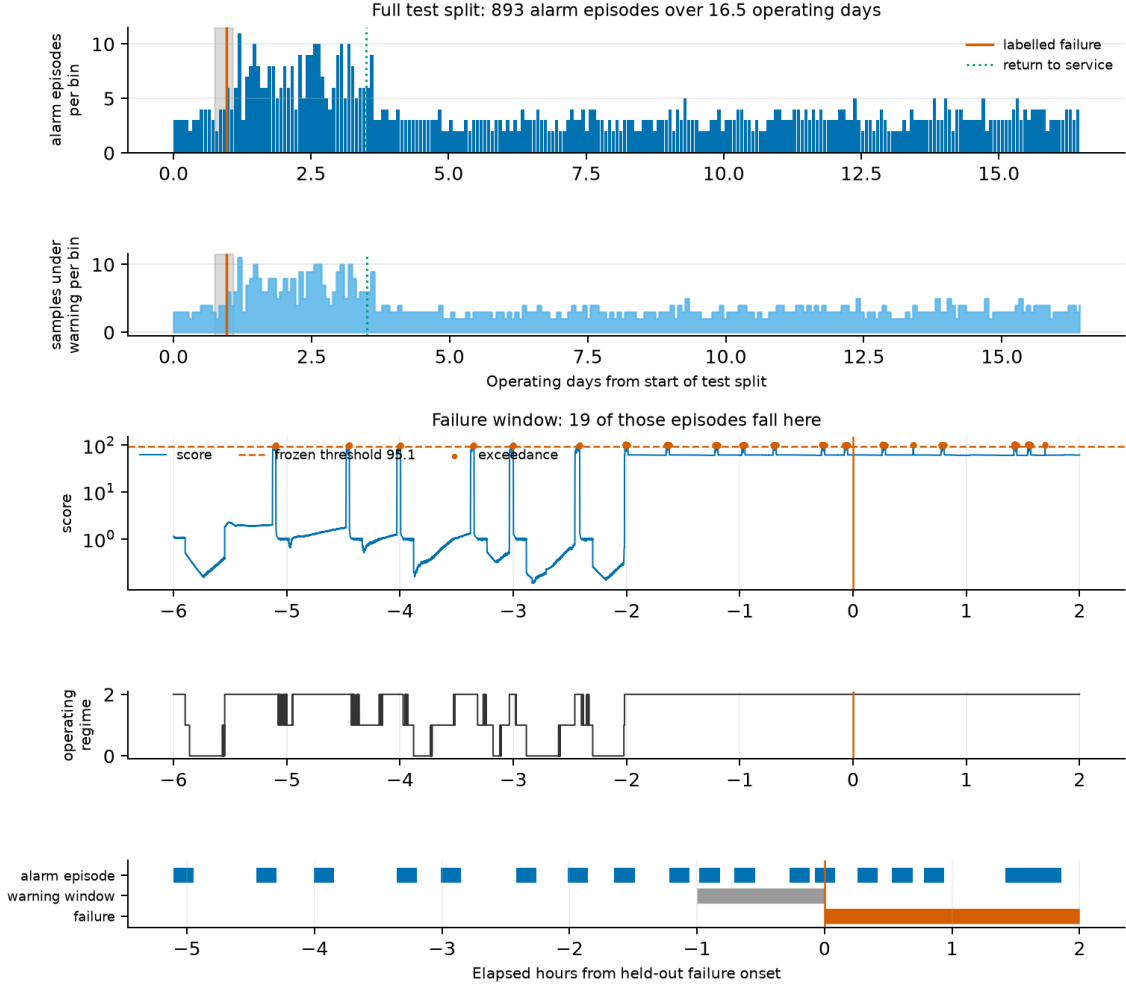


Figure 9: Global and local views of the MetroPT2 held-out failure `metropt2_test_failure_001` under the Dynamical EVT score method (`robust_max_abs_z` score). Threshold quantile 0.980 gives a frozen test threshold of 95.1; declustering run length 13 samples, merge gap 0 samples, warning horizon 1.0 h. The test split runs 16.5 operating days (1,423,388 samples). Globally the method raises 893 alarm episodes, of which 889 are false, at 53.96 false alarms per operating day; 19 episodes fall inside the plotted failure window and 15 of those are false. The matched alarm leads the failure by 0.98 h. Time under warning is 893 samples (0.06% of the split). The only maintenance boundary the processed data records is the end of the single labelled failure interval.

8 Discussion

8.1 Estimator finite-sample validity

The Monte Carlo results support a bounded interpretation of the EVT diagnostics. Runs, Ferro-Segers, K-gaps, reciprocal mean cluster size, block clustering, and no-declustering references are useful for detecting estimator instability across process families, thresholds, missingness, and dynamical structure. They do not by themselves identify maintenance-relevant degradation. The practical implication is that recurrence estimates should be treated as conditional diagnostics tied to the chosen observable, threshold, run length, and sampling process.

8.2 Target-region transferability

The central empirical finding is directional. Under the seven-feature construction the failure-prototype region missed the held-out MetroPT failure while detecting the MetroPT2 failure with 157 alarm episodes and 9.41 false alarms per operating day, materially better than the registered recurrence score on MetroPT2. A genuine transfer experiment changes how that should be read. On the five schema-compatible features the region fitted on MetroPT transfers to MetroPT2 and outperforms a region fitted on MetroPT2 itself, while the reverse direction fails outright, and the original MetroPT non-detection turns out to be a property of including two channels whose distributions do not overlap between the datasets.

Two cautions follow. The first is that a recalibration step cannot be assumed benign: recalibrating the threshold or the scaling on the destination rescued the failing direction and degraded the working one, so recalibration should be reported as a separate experiment rather than folded into a single transfer number. The second is that a feature set chosen for within-dataset performance is not automatically a feature set that supports transfer, and here it actively prevented detection. The evidence supports directional and feature-dependent transfer, not a general target-region solution, and two held-out failures cannot settle the question either way.

8.3 Score, threshold, clustering, and alarm-conversion failure

The failure mode is layered, and the decomposition locates each method’s layer rather than assigning one explanation to a dataset. Twenty-five of the 38 rows reach the failure and fail on alarm burden, nine fail at the score layer, two at event matching, one at the threshold, and one is not estimable. The concentration after alarm conversion is the substantive finding: for most of these methods the observable carries pre-onset ordering and the conversion from exceedances to alarms destroys its operational value.

The route to that finding is worth recording, because it bears on how much weight a layer diagnosis can carry. An earlier version of this work reported MetroPT as a score-layer failure on a separation of 0.45. That separation was produced by a scaling denominator with no lower bound, under which a channel that idles near zero acquired an interquartile range of 0.004 and dominated a maximum-absolute- z score. The diagnosis was correct about the layer in a narrow sense, since the score was indeed where the information was lost, but wrong about the cause and therefore about the repair: the fault was in preprocessing, not in the observable, and flooring the denominator recovers a separation of 0.78. A layered diagnosis inherits every assumption of the pipeline it inspects, and a score-layer verdict should be read as a prompt to audit score construction before concluding that an observable is uninformative.

What survives the correction is the alarm-conversion result. The registered score reaches both failures and needs thousands of episodes to do it, and the circular-shift null shows that a detector alarming that often matches essentially regardless of whether it anticipated anything. Recall at these rates is close to uninformative, which is why the null belongs beside it.

Two cautions. The separation is measured against one failure window per dataset, so it describes these events rather than estimating a population quantity, and the confidence column

in Table 6 records that. And a low separation under the declared one-hour horizon does not exclude separation at a different lead time, which is the competing explanation the table names for every score-failure row.

Two cautions follow from reading the stages as a flow. The counts change unit at every stage, so a steep decline between exceedances and episodes reflects a change from samples to episodes as much as any reduction the method achieved, and comparing methods across stages is not meaningful. More importantly, the labelled failures are not the end of the detector’s flow. Placing them there, as an earlier version of this figure did, invites reading the decomposition as a derivation from failures back to failures; they are a separate track that meets the detector only at the matching step, and the pipeline never uses them before that point.

8.4 Comparison With Baselines

Simple baselines remain competitive under the held-out event policy, in the weak sense that they fail in the same place. The engineering threshold matched both compressor events, but on MetroPT it does so with a separation of 0.45, which is the same score-layer failure as the registered method rather than a competitive detection. Its burden relative to the registered score depends on the axis measured: fewer alarm episodes, but far greater warning exposure. That asymmetry is itself the operational point. A method can be made to look better on episode counts by merging its alarms into fewer, longer blocks, which is not an improvement an operator would recognise, so claiming a burden advantage requires naming the burden. Classical EVT and anomaly baselines did not remove the basic precision problem. The proposed recurrence score therefore adds diagnostic structure, but it does not add measurable operational value on these held-out events.

8.5 Matched Negative Controls

Matched controls prevent over-interpreting one favourable target-region result, but only when they are conditioned on detection. Comparing alarm burden without that condition rewards a region for not detecting: a region raising no alarms attains the best possible burden and the worst possible outcome simultaneously. Both compressor results change once the condition is imposed.

On MetroPT the failure-prototype region raised no alarms and missed the failure, and so did every one of the 3000 control draws across six families. The comparison is therefore vacuous: a zero burden that is matched by every control and accompanied by no detection is not evidence of anything. On MetroPT2 the region detects, and conditioning lowers its standing against every control family. In two of the six it falls below the control median, and those two are the informative ones: occupancy-matched random regions and prototype-source permutations preserve the region’s own structure while breaking its relation to the failure. The families the region still beats are those whose draws detect almost always, where detection separates nothing.

The honest reading is narrower than the earlier one. Against chance regions that share its structure and also detect, the learned region is not better; against controls that detect trivially it is. Part of what the unconditioned comparison measured was the difficulty of the detection problem rather than the quality of the region, and the residual advantage rests on control families that discriminate least.

8.6 Independent-Event Scarcity

Only two independent held-out compressor failures are available. This prevents stable estimation of event-level uncertainty, calibration, and failure-type heterogeneity. Timestamp counts cannot substitute for independent failures because alarm episodes are temporally dependent and can cluster around the same operating regime.

8.7 Operational Implications

The workflow is useful as an audit of how extreme-score methods fail. It exposes whether apparent score excursions survive thresholding, declustering, alarm merging, and event matching. It does not establish deployable early warning. Deployment would require prospectively validated target regions, enough independent failures or entities, acceptable false-alarm burden, useful lead time, and comparison against simple policies under the same episode-level evaluation.

9 Conclusion

This study demonstrated a reproducible way to test whether clustered extremes become early warnings, and the answer on the evaluated public data is negative. What is supported is the diagnostic apparatus: the Monte Carlo study establishes where the extremal-index estimators are usable and where their ranking inverts, and the decomposition identifies which stage of each method destroys the signal. What is not supported is early warning. The registered recurrence score matched the two held-out compressor failures, but only with 2631 and 852 alarm episodes and false-alarm burdens of 105.34 and 51.47 alarms per operating day, one to two orders of magnitude above what alarm-management practice treats as absorbable. On MetroPT the match is not even evidence of detection: the score ranks the pre-onset window below chance, and a detector alarming that often matches by coincidence with probability 0.99. Nor is the comparison against simple policies settled by episode counts alone: the engineering threshold raises fewer episodes while keeping the system under warning for most of the MetroPT test period, so alarm burden is only interpretable when interruption frequency and warning exposure are reported together.

Target regions do transfer, but directionally and only on features that support it. Two of the seven candidate channels have no distributional overlap between the datasets; on the remaining five, a region fitted on MetroPT outperforms on MetroPT2 a region fitted on MetroPT2 itself, while the reverse direction admits ordinary operation half the time. Recalibration rescues the failing direction and degrades the working one, so transfer cannot be reported as a single number. The same feature restriction, not the method, explains the region’s MetroPT non-detection.

The observed MetroPT2 region does not exceed matched controls once the comparison is conditioned on detection. Against occupancy-matched random regions it falls from the 65th percentile of joint utility to the 28th, and against prototype-source permutations from the 66th to the 27th; the two families it still beats are those whose draws detect almost always, where detection separates nothing. On MetroPT its zero burden is vacuous, since neither it nor any of the 3000 control draws detected the failure. No evaluated method is operationally plausible on either dataset.

The main failure mechanisms are separated per method rather than per dataset. Of the 38 dataset-method rows, 25 fail after alarm conversion, nine at the score layer, two at event matching and one at the threshold. For most methods on both datasets the observable ranks the pre-onset window above chance and the conversion to alarms is what removes the value, which is the layer a future method could plausibly address. The failure-prototype region on MetroPT remains an exception in the other direction, admitting no exceedance at all, and the SPOT-style baseline ranks the window correctly while raising its alarms elsewhere. Non-event datasets add stress-test evidence only; their cycles, vehicles, and wafers are not independent compressor failure episodes.

Target-region methods add diagnostic value by exposing heterogeneity between MetroPT and MetroPT2, but no evaluated method establishes reliable operational early warning. The evidence required next is specific: more independent failure episodes, prospectively frozen target regions, stable regime definitions, deployment-realistic alarm policies, and external validation showing acceptable recall, false-alarm burden, lead time, and utility under the same event-level

protocol.

Code and Data Availability

Code, configuration files, data-acquisition scripts, generated-paper tooling, and reproducibility materials are maintained in the project repository [17]. It carries a semantic version tag, a BSD-3-Clause licence, `CITATION.cff` and `codemeta.json` metadata, release notes, and an artifact manifest recording the hash of every generated figure and table, so every number in this paper is traceable to the commit that produced it.

The repository is public, and the results in this paper correspond to the tagged release `v0.3.2`. A reader can retrieve that state, install the pinned environment from the lock file, and re-run the pipeline that produced every figure and table reported here.

That release is archived, and the software citation resolves to an immutable deposit rather than to a mutable host. The cited identifier is the concept DOI [10.5281/zenodo.21803768](https://doi.org/10.5281/zenodo.21803768), which names the software across all its versions. Every number reported in this paper was produced by release `v0.3.2`, and that version string is what pins the state rather than the DOI itself; readers who want an identifier for that snapshot alone can use its version DOI, [10.5281/zenodo.21803769](https://doi.org/10.5281/zenodo.21803769).

Citing the concept DOI is deliberate. A version DOI is minted from a snapshot after that snapshot is frozen, so an archive can never contain the identifier issued for it, and a paper pinned to one is pinned to an archive that does not describe itself. A concept DOI is fixed once and does not change, so it remains correct as the software continues, and naming the version in the text supplies the precision the identifier gives up.

Raw third-party datasets are not redistributed by this repository; the acquisition pipeline stores them locally under `data/raw/` and documents their public sources through the dataset registry.

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